Predicting Where People Look at Ads: An Open Benchmark and a Compact Attention Model Fine-Tuned on Design Eye Tracking

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Code, model and data pointers: https://github.com/pranikz/MediaAttention

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Abstract

Predicted-attention tools are widely used to pre-test advertising, yet their accuracy claims are hard to compare and the best open models were trained on natural photographs rather than designed content. We make three contributions. First, we review 15 commercial predictive-attention tools from public sources and show that headline claims of "92–95% accuracy" are mostly AUC expressed as a share of the inter-observer ceiling, a saturated metric on which a centre-only prior already scores 0.78 on MIT300. Second, we benchmark six openly available models, from a 3.7M-parameter network to the 452M-parameter DeepGaze MSDB, against human eye tracking on webpages (FiWI) and on 1,980 posters, app screens and webpages (UEyes), none of which they were trained on. On design content, model size does not buy accuracy: DeepGaze MSDB reaches a correlation (CC) of 0.485 with human fixation maps on UEyes, tied with DeepGaze IIE and below the 3.7M-parameter UNISAL (0.502). Third, we fine-tune UNISAL on the UEyes training split. The resulting model, MediaAttention-Ads v1, reaches CC 0.735 on the held-out UEyes test split and 0.752 on held-out posters, against 0.514 and 0.496 for the best other open model, with non-overlapping 95% confidence intervals, while also improving on out-of-domain webpages (0.631 vs. 0.616). The model runs in 0.04 s per image on a laptop. We release the model under Apache-2.0 together with MediaAttention, an open toolkit that turns attention maps into advertising metrics, and an open browser-based eye-tracking study kit.

1 Introduction

Attention is the first gate in advertising: an element that is not looked at cannot persuade or be remembered. Eye-tracking studies of print advertising show that pictures, text size and brand placement shape where attention goes and how it transfers between elements [1, 2], and visual salience alone can bias quick consumer choices [3]. Measuring attention with eye tracking is accurate but slow and costly, so a market has grown around *predicting* it: a creative is uploaded and a heatmap and scores come back.

Two problems limit this practice. Commercial tools quote accuracy on different metrics, test sets and viewing durations, often on their own data, so their claims cannot be compared. And the open models that lead public benchmarks were trained mostly on natural photographs (SALICON [4], MIT1003 [5]), while advertising is dense with type, logos and layout. This paper contributes:

1. A review of 15 commercial predictive-attention tools that decodes what their accuracy claims measure (Section 3).

2. An open benchmark of six models against human eye tracking on webpages, posters and user interfaces none of them were trained on, including a full run of the 452M-parameter DeepGaze MSDB (Section 5).

3. MediaAttention-Ads v1, a 3.7M-parameter attention model fine-tuned on design eye tracking that leads every open model on posters, ads and interfaces by a wide margin (Sections 4 and 5).

4. MediaAttention, an open-source toolkit that turns attention maps into advertising metrics, and an open browser study kit for collecting eye-tracking and implicit-response data (Section 4).

2 Background and related work

Attention prediction. Computational saliency began with feature-contrast models [6]. Since 2015, deep networks trained on large fixation datasets dominate. DeepGaze IIE [7] ensembles four ImageNet backbones; DeepGaze MSDB [8] combines CLIP and DINOv2 features at several scales with dataset-bias parameters; TranSalNet [9] adds a transformer encoder to a DenseNet; UNISAL [10] is a compact network for both images and video. The MIT/Tübingen benchmark [11, 12] evaluates models on held-out data and is the closest the field has to a neutral scoreboard.

Design and interface saliency. Design-oriented eye-tracking data is rare. FiWI [13] covers 149 webpages; UEyes [14] covers 1,980 posters, webpages, desktop and mobile interfaces viewed by 62 participants. SUM [15] trains on several datasets including UEyes. For video advertising, recent work finds video models far ahead of frame-by-frame image models on TV ads [16].

Metrics. No single metric captures prediction quality [17]. AUC measures ranking and saturates; shuffled AUC (sAUC) cancels centre bias [18]; NSS, CC, SIM and KL divergence compare maps. We use CC, the Pearson correlation between predicted and blurred human fixation maps, as the headline metric and report the rest.

3 What commercial accuracy claims measure

We surveyed vendor websites, documentation and white papers in October 2026. Where a claim can be decoded, it is usually AUC divided by the inter-observer ceiling: one vendor's "92.5%" is its MIT300 AUC (0.864) divided by the gold standard (0.934). On MIT300 every listed model scores between 0.84 and 0.89 AUC and a centre-only prior already reaches 0.78, so this ratio makes small differences look large. On CC the same tools separate clearly (Table 1). Only four commercial tools have third-party-computed MIT300 entries, and all rank below the best open models. The most detailed public technical document we found reports CC 0.944 on held-out data from the vendor's own ad corpus [19]; it cannot be compared with MIT300 scores, which use a different test set, ceiling and viewing duration. We found no external validation of secondary scores (clarity, memory, emotion).

4 MediaAttention

MediaAttention turns one prediction, where people will look, into the scores an advertiser acts on (Figure 1): an attention model predicts a fixation-density map, layout detection finds the elements that matter, and the two are combined into metrics and an Impact Score calibrated on the user's own ads.

Model (MIT300)	AUC	sAUC	NSS	CC	KLD	Type
DeepGaze MSDB	0.894	0.816	2.74	0.883	0.254	open, research
DeepGaze IIE	0.883	0.794	2.53	0.824	0.347	open, research
TranSalNet-Dense	0.873	0.747	2.41	0.807	1.014	open (MIT)
UNISAL	0.877	0.784	2.37	0.785	0.415	open (Apache-2.0)
Attention Insight	0.864	0.745	2.18	0.758	0.887	commercial
Emotiva	0.864	0.770	2.20	0.743	0.496	commercial
MedullEyes-V1	0.841	0.724	1.89	0.670	0.659	commercial
Clueify	0.881	0.765	1.49	0.575	0.889	commercial
Centre bias	0.783	–	1.10	0.446	0.951	baseline
Gold standard (humans)	0.934	0.883	3.14	0.983	0.060	ceiling

Table 1: MIT/Tübingen MIT300 leaderboard [12], accessed 21 October 2026. Clueify ranks well on AUC (0.881) but lowest among the tools on CC (0.575).

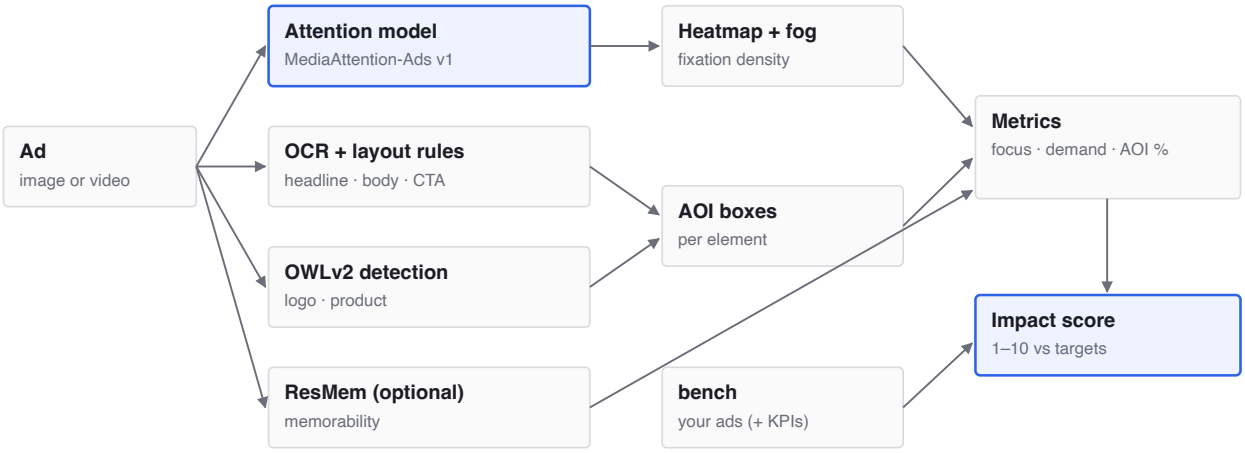


Figure 1: The MediaAttention pipeline for one image. Video runs the same steps per frame, adds cut detection and tracks areas of interest (AOIs) through each shot.

4.1 Attention network

We build on UNISAL [10] (Figure 2). A MobileNetV2 [20] encoder (2.2M parameters) extracts features at 1/8, 1/16 and 1/32 scale. Sixteen learned Gaussian prior maps are concatenated to the deepest features, letting the network learn content-independent viewing biases. A decoder with skip connections upsamples the result; a domain-specific head applies a 1×1 adaptation, resizes to the input and applies a learned 41×41 smoothing; a log-softmax produces a probability map over pixels. For video, a convolutional GRU on a bypass between encoder and decoder carries state across frames; images skip it. Batch normalisation, priors, adaptation and smoothing have one copy per data domain. We chose UNISAL as the base because it is the most accurate model with a permissive licence in our webpage benchmark and is 17–70× faster than DeepGaze, small enough to fine-tune on a laptop.

4.2 Fine-tuning on design eye tracking

Data. We use the UEyes training split: 1,872 images (posters, mobile and desktop interfaces, webpages) with eye tracking from 62 participants viewing freely for up to 7 s on a Gazepoint GP3.

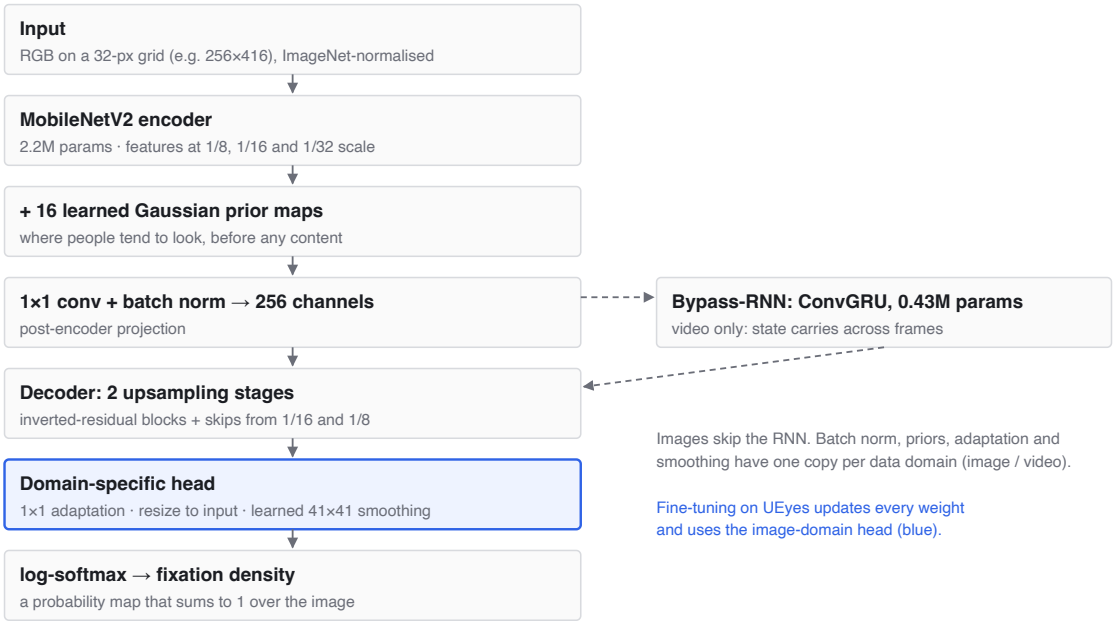


Figure 2: The attention network (UNISAL architecture), 3.7M parameters in total.

A random 10% (187 images) is held out to select the epoch. The UEyes test split (108 images, 27 per category) and FiWI are never used for training or model selection.

Targets. Fixations inside each image are placed on a map and blurred with a Gaussian of about one degree of visual angle (the dataset's 40 screen pixels per degree, converted to image pixels for each image's letterboxed display). The map is resized to a quarter of the network input resolution and normalised to sum to one.

Loss and optimisation. For predicted distribution P and target Q we minimise

$$\mathcal{L}(P, Q) = \mathrm{KL}(Q \,\|\, P) - \mathrm{CC}(P, Q), \tag{1}$$

the usual pairing for saliency training: KL divergence penalises missed attention, correlation rewards the overall shape. We start from UNISAL's SALICON weights and train all weights with AdamW (learning rate 3×10^{-5}, weight decay 10^{-4}), batch size one with batch-normalisation statistics frozen, for eight epochs (about 25 minutes on an Apple M3 Pro). The epoch with the best held-out CC is kept. Nearly all of the gain arrives in the first epoch and the curve then plateaus (Figure 3), which suggests more data, rather than more training, is the next lever. Only the image branch is tuned; video keeps UNISAL's DHF1K [21] weights, since no open ad-video eye-tracking dataset exists.

4.3 From attention maps to advertising metrics

Let p be the predicted map normalised to sum to one over N pixels.

- **Focus** (0–100). With pixels sorted by decreasing p, let k be the smallest count whose mass reaches 0.8. Focus is $100 \cdot \max(0,\, 1 - (k/N)/0.8)$: attention concentrated on a few elements scores high, evenly spread attention scores zero.

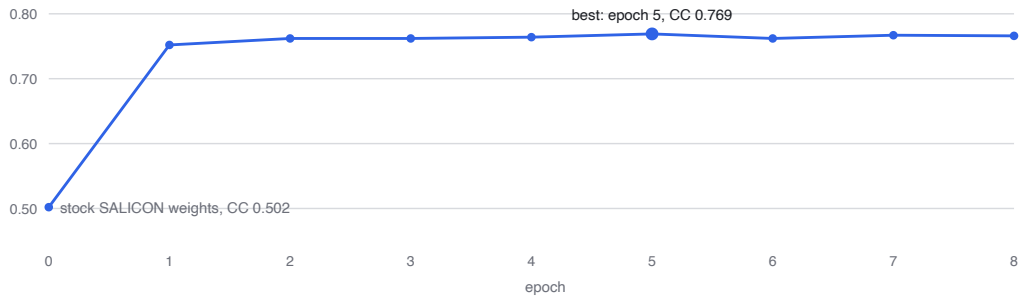


Figure 3: Held-out CC during fine-tuning (187 held-out training images). Epoch 0 is UNISAL's stock SALICON weights.

- **Cognitive demand** (0–100). The grayscale image is quantised to 32 levels and split into 16×16 blocks; the mean Shannon entropy [22] of the blocks is divided by $\log_2 32$, a simple clutter measure [23].
- **Areas of interest.** OCR (PP-OCR [24]) reads every text line. Lines within 75% of the largest text height form the headline; a short line (at most four words) on a uniform patch covering at most 10% of the ad is the call to action, and its box grows to the button's edges; the remaining text is body copy. OWLv2 [25] detects the logo and the product. AOI attention is $\sum_{x \in \text{box}} p(x)$; lift is attention share divided by area share.
- **Memory** (optional). ResMem [26], trained on LaMem [27] and MemCat [28], predicts recognition after a brief exposure [29].
- **Impact Score** (1–10). Each metric scores 10 inside its target range and falls linearly from 7 at the range edge to 1 at the end of the scale; the Impact Score is the mean. Targets are set from a folder of the user's own ads: the top 20% for higher-is-better metrics and the middle 50% for cognitive demand, or, given campaign KPIs, the middle 50% of what the best-performing fifth of ads scored.
- **Video.** A hard cut is a change of more than 40% in the grayscale histogram, at least 0.5 s after the previous cut; messages placed right after cuts are easily forgotten [30]. AOIs found at the start of each shot are tracked with pyramidal Lucas–Kanade optical flow [31, 32], and attention comes from the recurrent model fed at about six frames per second. Brand exposure over time follows [33].

4.4 An open study kit

Commercial tools rest on private panels. MediaAttention ships a static browser study instead: gaze is estimated from webcam face landmarks with a per-participant ridge calibration on a 9-point grid followed by an accuracy check; fixations are detected with a dispersion threshold; timed Yes/No word tests yield implicit scores weighted by each participant's relative response speed; and a recognition test against foil ads measures memory. Nothing is uploaded: participants download their own results file and choose whether to send it back. Webcam gaze is coarse (2–4 degrees versus about 0.5 for lab trackers) but, aggregated over participants, suffices to validate heatmaps on one's own creatives and is a route to openly licensed ad eye-tracking data.

Model	CC ↑	NSS ↑	sAUC ↑	SIM ↑	KLD ↓	pictorial CC	mixed CC	text CC
DeepGaze MSDB	0.676	1.54	0.748	0.642	0.462	0.751	0.686	0.592
DeepGaze IIE	0.649	1.50	0.729	0.629	0.504	0.726	0.680	0.541
MediaAttention-Ads v1 (ours)	**0.631**	**1.27**	**0.685**	**0.617**	**0.500**	**0.653**	**0.643**	**0.597**
UNISAL (stock)	0.616	1.31	0.661	0.613	0.575	0.676	0.641	0.531
TranSalNet-Dense	0.608	1.25	0.646	0.605	0.603	0.650	0.642	0.532
Centre-bias baseline	0.345	0.62	0.500	0.476	0.899	0.305	0.373	0.358
Spectral residual	0.209	0.40	0.562	0.449	1.052	0.220	0.249	0.159

Table 2: FiWI, 149 webpages; mean per-image scores, CC by page type on the right. No model was trained on FiWI.

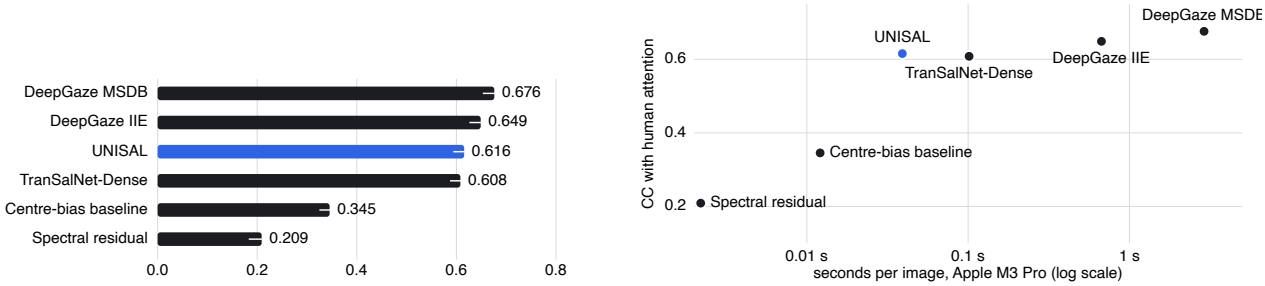


Figure 4: FiWI. Left: CC with 95% confidence intervals. Right: accuracy against inference time per image.

5 Benchmark

5.1 Setup

Data. FiWI [13]: 149 webpages at 1360×768, 11 viewers, 5 s free viewing, ground truth blurred with the dataset's own Gaussian ($\sigma = 32$ px). UEyes [14]: 1,980 images in four categories; fixations are mapped to image pixels as in the dataset's own code and blurred with about one degree.

Models. DeepGaze IIE and DeepGaze MSDB with a Gaussian centre-bias prior (MSDB with averaged dataset parameters); TranSalNet-Dense with SALICON weights; UNISAL with SALICON weights; a Gaussian centre prior; and spectral residual [34]. All are run as released, without fine-tuning, except MediaAttention-Ads v1.

Metrics and hardware. pysaliency [11] computes AUC, sAUC, NSS, CC, SIM and KLD per image; we average over images and report 95% confidence intervals as ± 1.96 standard errors. Timing is on an Apple M3 Pro (PyTorch MPS) after model loading. The full-UEyes DeepGaze MSDB run used a Colab T4 GPU with the same code.

5.2 Webpages (FiWI)

DeepGaze MSDB leads on every metric (Table 2, Figure 4). Every model is weakest on text-heavy pages: stock UNISAL scores 0.531 on text pages against 0.676 on pictorial ones. Fine-tuning on design content raises UNISAL's text-page CC to 0.597, the best of all models, and its overall CC from 0.616 to 0.631, although FiWI is out of domain for both.

Model	CC ↑	NSS ↑	sAUC ↑	SIM ↑	KLD ↓	poster CC	mobile CC	desktop CC	web CC
TranSalNet-Dense	0.534	1.01	0.655	0.586	0.700	0.523	0.586	0.543	0.483
UNISAL (stock)	0.502	0.97	0.656	0.573	0.779	0.494	0.521	0.526	0.468
DeepGaze MSDB	0.485	0.99	0.704	0.525	0.921	0.463	0.516	0.517	0.443
DeepGaze IIE	0.485	0.95	0.684	0.548	0.777	0.483	0.478	0.513	0.465
Spectral residual	0.232	0.45	0.623	0.463	1.006	0.257	0.198	0.290	0.185
Centre-bias baseline	0.220	0.37	0.497	0.468	0.963	0.257	0.201	0.198	0.225

Table 3: All 1,980 UEyes images, models as released. MediaAttention-Ads v1 is excluded because it was trained on the training split; see Tables 4 and 5.

Model	CC ↑	NSS ↑	sAUC ↑	SIM ↑	KLD ↓	poster CC	mobile CC	desktop CC	web CC
MediaAttention-Ads v1 (ours)	**0.735**	**1.53**	**0.731**	**0.680**	**0.399**	**0.752**	**0.820**	**0.692**	**0.674**
TranSalNet-Dense	0.514	1.00	0.646	0.559	0.771	0.496	0.601	0.482	0.476
UNISAL (stock)	0.467	0.92	0.640	0.543	0.864	0.454	0.504	0.456	0.452
DeepGaze IIE	0.454	0.92	0.667	0.522	0.871	0.458	0.460	0.453	0.445
DeepGaze MSDB	0.452	0.94	0.690	0.499	1.039	0.415	0.530	0.465	0.397
Centre-bias baseline	0.219	0.40	0.499	0.447	1.021	0.281	0.189	0.190	0.215
Spectral residual	0.214	0.44	0.616	0.433	1.125	0.285	0.131	0.270	0.169

Table 4: UEyes test split, 108 images (27 per category) never used for training or model selection.

5.3 Posters and interfaces (UEyes)

Table 3 runs the released models on all 1,980 UEyes images. Every model is far weaker than on webpages, and size does not help: the 452M-parameter DeepGaze MSDB ties the 104M-parameter DeepGaze IIE (CC 0.485) and trails both TranSalNet (0.534) and the 3.7M-parameter UNISAL (0.502). MSDB does lead on sAUC, which removes centre bias, so its ranking of salient regions is competitive; it is the shape of its maps that fits designed content poorly.

On the held-out test split (Table 4), MediaAttention-Ads v1 reaches CC 0.735 against 0.514 for the best other model, and leads in every category by 0.19 to 0.26. On posters, the category closest to print and display advertising, it reaches 0.752 [0.695, 0.809] against 0.496 [0.440, 0.551] for TranSalNet (Table 5); the intervals do not overlap.

5.4 Architectures compared

Table 6 sets the models side by side: UNISAL (Figure 2); TranSalNet, a DenseNet-161 CNN with a transformer encoder and CNN decoder; DeepGaze IIE, an ensemble of four ImageNet backbones with a readout network; DeepGaze MSDB, CLIP RN50x64 and DINOv2 ViT-B/14 features at several scales with dataset-bias parameters; and, for reference, the commercial Neurons model, a CNN encoder–decoder with ConvNeXt and ResNet-50 components trained on proprietary eye tracking from about 20,000 participants [19]. MediaAttention-Ads v1 is 20–120× smaller and 17–70× faster than the DeepGaze models, handles images and video in one network, and is the only accurate model whose code and weights carry a permissive licence; the DeepGaze repositories have no licence file. Its weakness is photo-led content: DeepGaze MSDB remains about 0.05 CC ahead on FiWI and further ahead on pictorial pages.

6 Discussion and limitations

What predicted attention can tell an advertiser. Models trained on free viewing predict bottom-up attention in the first seconds: what pops out. They do not model goal-directed search,

Model	CC ↑	95% CI	NSS ↑	KLD ↓
MediaAttention-Ads v1 (ours)	**0.752**	**[0.695, 0.809]**	**1.55**	**0.354**
TranSalNet-Dense	0.496	[0.440, 0.551]	0.98	0.792
DeepGaze IIE	0.458	[0.392, 0.524]	0.90	0.894
UNISAL (stock)	0.454	[0.389, 0.518]	0.90	0.913
DeepGaze MSDB	0.416	[0.351, 0.481]	0.86	1.079
Spectral residual	0.285	[0.209, 0.361]	0.58	1.082
Centre-bias baseline	0.281	[0.218, 0.344]	0.55	0.949

Table 5: UEyes test posters ($n = 27$), with 95% confidence intervals over images.

Model	Params	Video	Licence	s / image	UEyes test CC	FiWI CC
MediaAttention-Ads v1 (ours)	**3.7M**	**yes (recurrent)**	**Apache-2.0**	**0.04**	**0.735**	**0.631**
UNISAL (stock)	3.7M	yes (recurrent)	Apache-2.0	0.04	0.467	0.616
TranSalNet-Dense	76.6M	no	MIT	0.10	0.514	0.608
DeepGaze IIE	104.0M	no	none	0.67	0.454	0.649
DeepGaze MSDB	451.6M	no	none	2.90	0.452	0.676
Neurons (commercial)	n/d	separate model	closed	–	n/a	n/a

Table 6: Size, video support, licence, speed (Apple M3 Pro) and accuracy (CC). Parameter counts were measured by loading each model. The commercial model is described from its public tech paper [19] and could not be run.

audio, or the placement context of an ad. Predicted attention is a fast first filter for comparing creative variants, not a replacement for testing with people when stakes are high. Buyers should ask vendors for CC and NSS on held-out, third-party-scored data rather than a bare percentage.

Limitations. FiWI is small (149 pages, 11 viewers). UEyes is one lab panel of 62 participants viewing freely for up to 7 s; neither measures a one-second glance in a scrolling feed, and cultural coverage is narrow. Our model's video branch is not ad-tuned or validated on ad eye tracking, and the secondary metrics (focus, cognitive demand, memory) have not been validated against campaign outcomes. Commercial tools could not be run on the same data.

Future work. Tuning the recurrent branch on ad video, scoring creatives in feed context with short exposures, distilling a larger model on photographs while keeping the design data, and growing an open ad eye-tracking dataset with the study kit.

7 Conclusion

Commercial attention-prediction claims are rarely comparable, and where third-party scores exist, the best open models match or beat them. On designed content, larger models do not help; training on design eye tracking does. A 3.7M-parameter network fine-tuned on UEyes for 25 minutes on a laptop leads every open model on posters, ads and interfaces by more than 0.2 CC and runs in tens of milliseconds. The model, toolkit, benchmark and study kit are open.

Availability. Code (Apache-2.0), benchmark scripts, per-image results and the study kit: `https://github.com/pranikz/MediaAttention` [35]. Model weights (Apache-2.0): `https://huggingface.co/pranikz/MediaAttention-Ads`. MediaAttention-Ads v1 is a fine-tune of UNISAL (Apache-2.0) on UEyes (CC BY 4.0); no UEyes images are redistributed.

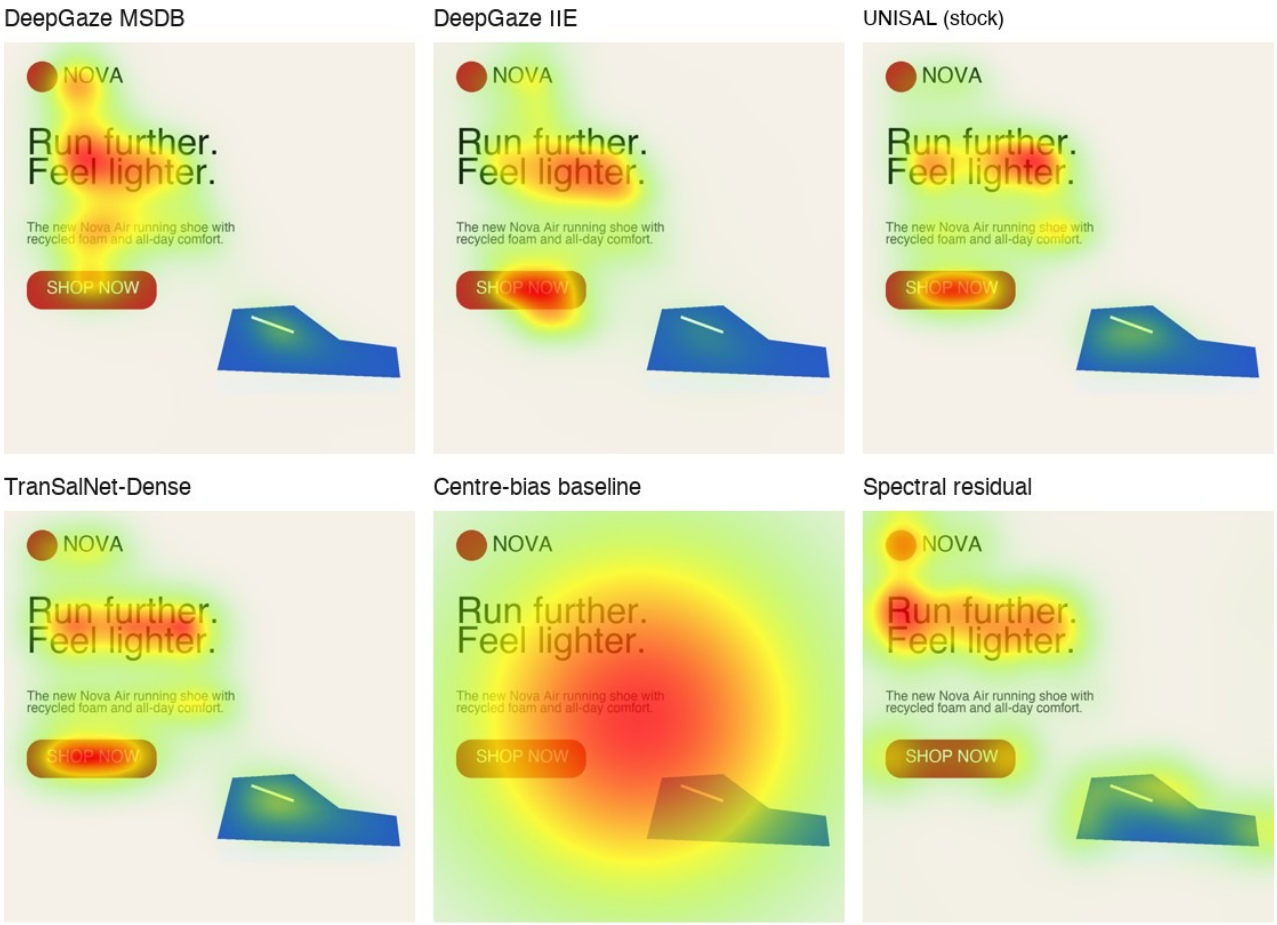


Figure 5: Predictions of the benchmarked models on the same synthetic ad (illustrative; no ground truth).

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